

Something Similar: Exploring the Usefulness of an On-the-Spot Meal Recommendation System for Health Goal Attainment

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Abstract

Recommendation systems for nutrition management have become increasingly popular. Current solutions focus on optimizing for users taste preferences, often neglecting practical factors including personal priorities, existing plans, and situational constraints. We propose *similarity-based* meal recommendation – a novel approach that considers users existing meal plans and recommends similar meals that also align with their nutrition goals. This approach is designed to support meal decisions in on-the-spot, particularly when individuals are in constrained settings with limited options for adjustment (e.g. a cafeteria). We conducted a feasibility study with 15 participants to evaluate the utility of similarity-based recommendations in constrained settings and refine the system design. Findings show (1) a novel approach for similarity-based meal recommendations, (2) insights into the dimensions of similarity influencing actionability across diverse, and (3) opportunities to enhance user control over the system’s design. The study contributes to the advancement of human-centered AI technologies in nutrition.

CCS Concepts

• Human-centered Computing; • User Studies; • User centered design;

Keywords

Nutrition Management, RecSys, User Studies, Mixed-Methods, Human-Centered AI

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1 Introduction

Food and nutrition are cornerstones of our cultural identities and daily routines. They also play a critical role in self-management of health, particularly for individuals with chronic diseases such as diabetes and heart disease [48]. However, managing nutrition is a challenging undertaking. Physiological responses to food are highly individual and complex, making it difficult to identify personally meaningful nutritional targets [5, 28, 51]. Even with clear goals, aligning daily meals with these targets requires persistence, nutritional literacy, and sustained motivation [8, 11, 21, 35, 43]. People must navigate competing priorities, balance preferences with goals, and adapt to dynamic contexts. The diverse socioecological factors influencing meal decisions highlight the need for human-centered nutrition technologies tailored to users’ unique decision-making contexts [20, 33].

Within the HCI community several types of human-centered technologies have been developed to help people navigate nutrition management [29, 33]. *Personal Informatics (PI)* tools help users log and reflect on personal data to inform future health behaviors [14, 18, 25, 27]. These systems help users uncover patterns in their data, however they require users to have enough expertise to use this information to determine the appropriate action to take; past work has shown users often seek more explicit, actionable insights in real time [7, 13, 19, 22, 30, 39, 40]. *Personal coaching technologies* are another set of systems that can provide expert guidance in setting and achieving health goals proving valuable in chronic disease [31, 32, 41]. These systems have shown immense potential in supporting nutrition management by helping users set concrete

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goals, evaluate progress, and increase knowledge. However, individuals still struggle to implement nutritional goals in the moment [19, 30]. Given this challenge, *recommender systems (RecSys)* can offer direct support, generating specific suggestions tailored to user preferences and needs [1]. Existing nutrition RecSys largely focus on recipe recommendations and optimize for user preference [2, 45]. Though preference-driven recommendation algorithms are effective in domains like entertainment, nutrition decisions demand an approach which balances personal tastes and specific nutrition goals (e.g. choosing lean proteins, choosing whole grains, and managing portion size) often under dynamic constraints (e.g. location, convenience, cost) [38]. Additionally, users' ability to act on recommendations is limited by the constraints of the situational setting, particularly if they are making meal choices on-the-spot and cannot feasibly prepare a new recipe. Consequently, there remain gaps between users' in-the-moment needs when they make nutritional choices and assistance provided by traditional nutrition RecSys.

In this work, we propose an approach to providing nutritional recommendations on-the-spot using **meal similarity and goal alignment** as the main criteria for identifying appropriate recommendations. In this approach, a nutrition RecSys takes as an input a meal the user is considering, such as a meal they plan to order for take-out (reference meal), assesses how the reference meal aligns with the user's nutrition goal, and if it does not align, recommends a meal *similar* to the reference meal that *does* align with the nutrition goal. We propose this approach can have unique benefits in constrained settings (e.g., a cafeteria) where users' choices are limited. Through a lab feasibility study with 15 participants, we examine (1) the utility of similarity-based meal recommendation in such constrained settings and (2) identify other factors that influence user experience. The study showed that in constrained settings users indeed preferred recommendations similar to meals they were already contemplating. However, the study also revealed that similarity is a complex and multifaceted construct and further research is needed to more fully understand its dimensions. This study makes the following contributions:

1. **A similarity-based recommendation:** We propose a novel recommendation approach to support nutrition decision-making in highly constrained settings, by generating suggestions similar to users existing plans.
2. **Nuances in recommendation similarity:** We investigate dimensions of similarity which influence the perceived actionability of recommendations, offering insights with implications beyond nutrition and health.
3. **Enhancing in-the-moment control:** We discuss enhancing user control over the system, algorithmic mechanisms, and output presentation, contributing to the broader design of human-centered technologies.

2 Proposed solution: recommending similar & goal-aligned meals On-the-Spot

We propose a *similarity-based approach to designing nutritional recommendation systems*. Our approach starts with a user logging a meal they plan to purchase or consume, which likely reflects their preferences and situational factors. Once the meal is captured, the RecSys determines whether this meal meets an

individual's nutritional goal or not. Past work in the field has explored computational methods to help users identify if a planned meal meets a given nutrition goal using self-tracked meal data [19]. However, if their planned meal does not meet their health goal it remains unclear how to suggest actionable adjustments to users. We propose recommending alternative meals that are similar to the planned meal but satisfy the nutrition goal. By focusing on suggesting meal similar to users' initial plan, this approach offers decision support without requiring data about user preferences and meal history, as other approaches demand. Key components of this approach are enumerated below and summarized in Figure 1.

- **User Inputs:** This system seeks two types of user input:
 - (1) **Nutrition Goal:** To guide the system, users specify their nutrition goal, such as: "Make half my meal non-starchy vegetables", "Choose lean proteins"). Health goals could be set in advance or selected by users on the spot.
 - (2) **User Meal Plan:** Users also capture a meal they considering purchasing, ordering, or consuming. This could be captured as free-form meal description or as structured text. In this study, we focus on free-form meal descriptions, because prior work in HCI has identified free-text as a low-burden way to capture data in-the-wild [16].
- **Meal Catalog:** Recommendations are generated from a diverse library of free-text meal descriptions labeled on their alignment with different nutritional goals. This library of meals could be collated from any free text meal datasets including standardized recipe data or crowdsourced meal records, enabling customization.
- Within the corpus, each meal is represented as a vector by using TF-IDF (term-frequency-inverse-document-frequency) to encode the meal title and description. TF-IDF down-weights tokens frequently used across the corpus, thereby encoding the uniqueness of tokens in one meal relative to others in the corpus [34].
- **Similarity-Based Recommendation Mechanism:** In order to generate goal-aligned recommendations, the meal catalog is first filtered to the subset of meals that meet the users chosen health goal to create a **goal-aligned meal catalog**. Then, the text similarity between the user's meal plan and each item in the **goal-aligned meal catalog** is computed and ranked, with the most similar meals returned as recommendations.
- To compute similarity we used *cosine similarity* which calculates the distance between two text vectors, in this case the user's meal plan and each meal record in the catalog [6]. Cosine similarity ratings range from 0 to 1, with ratings closer to 1 indicating greater similarity.
- **Output:** Finally, the system outputs a ranked list of the most similar meal to the user's initial plan, from all the meals in the catalog that meet the user's health goal.

3 Feasibility Study Methods

To evaluate the feasibility of the proposed approach, we conducted an experimental study that took the form of a 60-minute virtual individual research session. Participants were asked to choose a nutritional goal and then pretend that they were making meal

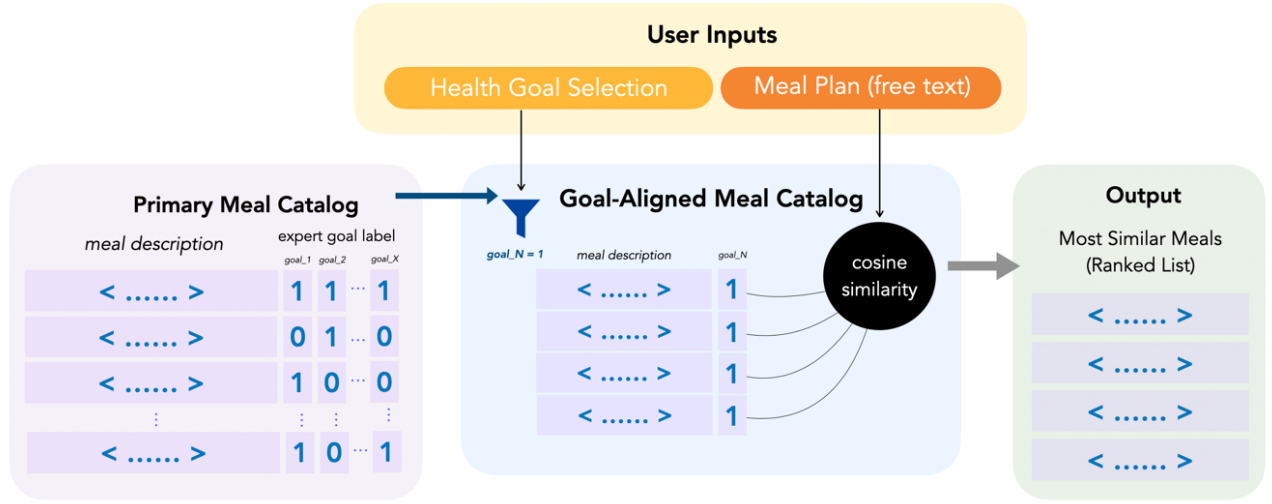


Figure 1: Diagram of Key Components of Similarity-Based Recommendation Approach

choices in the constrained setting of a cafeteria. Specifically, they were asked to make a choice from a menu that only included meals not aligned with the nutritional goal, then were presented with recommendations with varying degrees of similarity and asked to choose recommendations they preferred and were most likely to follow.

3.1 Study Design

In line with the proposed approach, participants selected one of two nutrition goals based on their personal aspirations: “Make half my meal non-starchy vegetables” or “Choose lean proteins.” Participants were shown a café menu and chose a meal they would like to order, referred to as the “reference meal”. The study was designed such that none of the possible reference meals met the participants chosen health goal. Participants were then presented with four recommendations that were all aligned with the chosen nutrition goal but varied in the degree of similarity to the reference meal (Figure 2).

For the user evaluation, recommendations were generated from a library of 3k meal records collected from our prior work [9, 16]. Each meal record contains a free-text meal title and meal description, and binary expert label for if the meal aligns with each nutrition goal (labels were generated by six expert Registered Dietitians using a unified protocol). The recommendation engine selected four meals from this corpus; two of the recommendations were similar to the anchor meal (cosine similarity >0.3), while the other two were dissimilar (cosine similarity <0.3).

Participants selected the recommendation they were most likely to follow, explained their choice, and reflected on their overall impressions of the system. We used a one-sample difference in proportions test to evaluate participants’ preference for similar recommendations ($H_0=0.5$). Sessions were transcribed and qualitatively analyzed using thematic analysis.

3.2 Participant Recruitment

We aimed to recruit participants from a variety of backgrounds, using a combination of online recruitment platforms such as Craigslist, Amazon Mechanical Turk (AMT), and a participant recruitment website hosted by [Institution]. Participants self-assessed eligibility using the following criteria: aged between 18 and 75 years old, no cognitive impairments, English fluency, and had access to the internet. Research sessions were conducted virtually via Zoom and participants were compensated (\$15). This study was reviewed by the Columbia University Medical Center IRB.

4 RESULTS

We summarize findings on participant backgrounds and their technology usage for meal decision-making (4.1); the utility of similarity-based meal recommendations in each scenario through quantitative (4.2.1) and qualitative analyses (4.2.2); and additional design considerations raised by participants (4.3).

4.1 Participant Background and Technology Experience

Participants ages ranged from 19 to 75 years of age; most participants identified as women (73%); just under half of identified as white (47%). 27% of participants self-identified as not having sufficient financial resources to feel comfortable (Table 1). Several participants noted that their eating practices were informed by religious beliefs (e.g. keeping Kosher, Halal) or pre-existing health conditions (e.g. pre-diabetes, Pott’s Disease).

All participants described using technology to support their health, such as watching cooking videos on YouTube and food, movement, or menstrual activity tracking apps. In daily life they also used several technologies that aided meal decision-making by helping to outsource cognitive burden and facilitate reflection and organization. For example, general purpose mobile apps such as “Notes” or store-specific apps were used to organize shopping lists

Reference Meal – Italian sandwich and bottle of water: Italian sandwich, ham, provolone, salami, capicola bologna, lettuce/tomato		
Library ID	Meal Description	Similarity
1092	Sandwich with Side Salad: Apple, lettuce salad, sandwich and an oatmeal cookie (add a side salad)	0.74
0307	Eggplant Marinara sandwich: eggplant, red pepper, lettuce, tomato, red onion, spinach, cucumber, provolone cheese, on herb bread with marinara	0.42
...		
1002	Chicken alfredo and broccoli and bottle of water: Chicken, broccoli, pasta with alfredo, water	0.18
0004	fruit and nuts: one apple, peanuts, chocolate covered almonds	0

It's lunchtime! What do you plan to eat?

Italian sandwich, ham, provolone, salami, capicola bologna, lettuce/tomato

This meal may not meet your goal to "Choose lean proteins in each meal". Here are some alternatives...

A. **Fruits and Nuts:** one apple, peanuts, chocolate covered almonds

B. **Eggplant Marinara sandwich:** eggplant, red pepper, lettuce, tomato, red onion, spinach, cucumber, provolone cheese, on herb bread with marinara.

C. **Chicken alfredo and broccoli and water:** Chicken, broccoli, pasta with alfredo, water

D. **Sandwich with Side Salad:** Apple, lettuce salad, sandwich and an oatmeal cookie

Figure 2: (a) Example of reference meal and cosine similarity ratings for four meals from the goal-aligned meal library; two similar to the reference meal (>0.3) and two dissimilar (<0.3) (b) An example of recommended meals as shown to users in feasibility study.

Table 1: Demographics of participants in the study (N=15)

Category	N (%)	Category	N (%)
Age		Race	
Mean (SD)	35.9 (18.6)	American Indian or Alaska Native	1 (6%)
Range	19 – 75 y.o.	Asian	3 (20%)
Gender	10 (67%)	Black or African American	4 (27%)
Woman	4 (27%)	White	7 (47%)
Man	1 (6%)	Ethnicity	3 (20%)
Another Term: Transgender Man	5 (33%)	Hispanic/Latinx	4 (27%)
Education	10 (67%)	Chronic Condition (any)	
Some College or less		Yes	
BA or higher			

and aggregate discount relevant coupons to use at checkout. Participants also regularly used technology to assist priority-aligned meal discovery, such as using food recommendation apps or websites to discover recipes that enabled them to balance high-level priorities such as nutrition goals, budget, taste, and preparation time. Several participants used generative AI tools like ChatGPT to support everyday tasks with P21 exploring it for meal planning: “My friend typed into ChatGPT a sample menu. We ended up not using it just because it sounded really in depth and it had like seven meals planned out for us but I think if I’m ever looking for something really specific...or when I’m really stuck on a recipe...I think it would maybe help.”

However, participants seldom described using technology to aid meal decision-making in constrained settings. The technologies they did describe were solely to enable instantaneous action, such as apps to place an order for food. As a result, in these settings participants had to rely on their own judgement to consider possible adjustments, try and weigh priorities, and navigate cognitively taxing decisions in-the-moment.

4.2 Decision-making Context and Recommendation Similarity Preference

We used a mixed-methods approach to assess the usefulness of **similarity-based meal recommendations** in supporting meal decision-making and discuss both quantitative (4.2.1) and qualitative findings (4.2.2).

4.2.1 Quantitative Findings. The difference in proportions test showed a significant preference for similarity-based meal recommendations ($z=3.35$, $p<0.001$). Out of 15 participants, 14 (93%) preferred recommendations highly similar to their initial plan.

4.2.2 Qualitative Findings. Qualitatively, participants reflected that in *constrained settings* recommendations served as an instruction for what they should eat; as a result, they sought recommendations that were specific and implementable in the moment. They were particularly interested in similarity-based recommendations because it included attributes aligned with what they were already planning to eat, and only required small adjustments to their existing plans. P20 described: “When I already have a meal in mind, I’m thinking about that meal. It’s unlikely I’m going to go past a certain level of similarity”.

However, the attributes that made a recommended meal similar or different from the reference meal in these constrained settings was multifaceted. We enumerate four main dimensions of similarity users discussed below:

- **Similarity in Meal Form:** Most commonly, participants discussed wanting a meal of the same *form* to what they already planned to eat, particularly if they were in the constrained setting. For example, they preferred to swap one sandwich for another, even it included different ingredients. Not surprisingly, these considerations were based on both taste, functionality, and the convenience of findings a substitute that was similar to the originally chosen meal. For example, participants remarked on waiting in a line for a particular type of food (“salad line/bar”, “sandwich station”) making alternatives that shared the same form easier to act on. P20 said: *“I’m picturing the stations in my head and thinking about which station I would go to. Pasta is a different station, and sandwich is the same station [as my initial choice]. But that’s also usually how I pick my food”*
- **Similarity in Meal Cuisine:** Participants also considered the similarity of the meal cuisine when considering recommended alternatives. They articulated that a given cuisine often had many meals with similar flavors to what they were in the “mood” to eat. On a practical note, they also pointed out restaurants were often cuisine-focused, making recommendations of similar cuisines not only appealing but more actionable.
- **Similarity in Meal Ingredient Preparation:** Somewhat unsurprisingly participants gravitated towards recommendations with similar ingredients to their initial plan. However, those options were less appealing if they contained other ingredients participants disliked or ingredients they had specific practices around, for example religious practices. Participants reflected that in less constrained circumstances they might be open to those recommendations, but when constrained this often deterred them from the option altogether. Practically, they also highlighted that ingredient preparation made certain kinds of modifications more feasible in the moment.
- **Similarity in Nutrition Value:** Another consideration participants described was the macronutrient content of the recommendation. This was particularly the case in *constrained* settings where participants were generally considering the recommendations as an alternative to their existing plan. They discussed ensuring the meal recommendation selected had a similar nutritional profile both quantitatively (for example equally protein rich) as well as qualitatively (the meal was equally heavy or filling) to their initial plan. P07 reflected: *“I would prefer the chicken marinara sandwich [or] Dagwood [sandwich] over the fruits. . .remember, we are talking about the day when I’m feeling hungry. . .Fruits are not going to quench that requirement”*.

4.3 Additional design considerations for meal recommendation systems in constrained settings

While participants reflected positively on the system and perceived it to be useful, particularly in constrained settings, they sought greater on-the-spot control at the level of the overall system, algorithm, and recommendation output.

First, at the level of the overall system, users wanted to be able to provide direct input about the overarching unchangeable assumptions the technology was making about them. They wanted ways to provide direct input around food allergies or dietary restrictions. They also wanted to turn on or off certain features across the system. For example, while many participants found value in viewing macronutrient information, one participant with a history of disordered eating worried this information might be triggering and wanted a way to turn off this feature. Users also discussed how their own technology needs evolved with time and experience, so while features such as explanations for why suggested meals met goals might be useful at the start, they wanted the option to turn off these features as they gained experience.

Second, at the level of the algorithm, participants expressed interest in being able to adjust the cosine similarity threshold to narrow or broaden of the degree of similarity between their original meal plan and recommendations in real-time. They sought ways to specify the dimension of similarity they were interested in to better “tune” the recommendation algorithm to meet their momentary needs.

Finally, when it came to system output, participants desired control over how recommendation results were displayed and the ability to manipulate the presentation of results in real-time to aid decision support. They suggested additional controls like filters that enabled them to sort recommendations by other data elements that aligned with their preferences such as cost, reviews or ratings to ascertain taste, or within recommendation ingredient substitutions to give them fine-grained control to integrate (rather than overlook) higher priorities in constrained settings.

5 Discusson

This work introduces a novel approach to delivering nutritional recommendations on-the-spot, emphasizing similarity and goal alignment with users’ meal plans and nutrition goals. A feasibility study with 15 participants assessed this approach in a simulated constrained cafeteria setting. They found **similarity-based recommendations** highly useful due to their alignment with existing plans and practical applicability in constrained settings. “Similarity” emerged as a multidimensional idea, encompassing dimensions not explicit in the meal descriptions text, such as: meal form, cuisine, ingredient preparation, and macronutrient composition. This suggests going beyond the lexical similarity to incorporate these dimensions to improve decision-making alignment. The study also highlights the importance of user control and fine-tuning on-the-spot, to enable better adaptation to dynamic situational factors. These findings offer valuable insights for designing nutrition technologies that align with users’ goals and support decision-making on-the-spot.

5.1 Utility of Similarity-Based Recommendation on-the-spot

Throughout the study, participants described how meal decisions in constrained settings were often automatic, leading them to prioritize convenience over long-term health goals. Decision science theories distinguish between two thinking modes: System 1, which is fast, intuitive, and automatic, and System 2, which is slower and more deliberate [23, 24]. In our study, participants discussed making meal choices in constrained settings as aligned with System 1 thinking, choosing options that met immediate needs but often overlooked broader health goals. Consequently, recommendations were perceived as supporting decision-making by serving as a source of instruction, reducing cognitive burden. This made similarity-based recommendations ideal as they closely aligned with meals that were accessible and aligned with users' momentary preferences so they could easily act without their health goals falling by the wayside.

While collaborative filtering and content-based algorithms dominate RecSys, this study demonstrates the feasibility of alternative mechanisms, such as similarity, for situationally appropriate recommendations. Similarity-based methods are already applied in commercial settings, such as online shopping to find related products [12]. In the HCI community, previous work has leveraged similarity to recommend recipes, images, and ingredient substitutions [4, 37, 44, 50]. For example, Yum-Me by Yang et al. computed image-based similarity to support exploration and discovery [50]. However, such systems still focus primarily on similarity to individuals' past meals to infer preferences. Here, we argue that similarity to a meal a person is already considering cooking, ordering, or purchasing could also help an individual meet nutrition goals and address situational constraints. Research in RecSys proposed different approaches to addressing such situational constraints. For instance, Viswanathan et al.'s Mindsets, a travel recommendation system, dynamically adapted suggestions based on user constraints, such as "refining plans" or making "on-the-spot" decisions [46].

Our study also suggests that similarity, particularly in reference to meals, is complex and multifaceted. Understanding the many dimensions of similarity can enable nutritional RecSys to better align with users' immediate goals, offering practical support in real-world decision-making contexts. Some past research has already tackled these questions. For instance, Pinxteren et al. identified dimensions such as macronutrients, cuisine, and ingredients in their taxonomy of recipe similarity, many of which align with the findings of this study [37]. Our findings expand on these findings by introducing a multidimensional view of similarity specific to priorities for constrained on-the-spot meal decisions, encompassing meal form, cuisine, ingredient preparation, and macronutrient composition.

5.2 Enabling Control: Mechanisms to Support User Autonomy

Beyond this emphasis on similarity as a basis for recommendations, our study highlighted the importance of including mechanisms for user control when building systems that use computational inferences to generate recommendations. These findings are consistent with prior literature that highlights the importance of autonomy

and control in shaping individuals' experiences and choices, particularly in the context of health self-management and patient empowerment [15, 36, 39, 49]. It is also one of the prevalent themes in the emerging body of literature on Human-Centered AI that increasingly highlights the importance of users' control and autonomy in their interaction with AI-powered interactive systems [10, 26, 42]. Our findings suggest that future nutritional recommendations systems should align with these prior efforts and strive to give users appropriate level of control that could enhance their sense of autonomy. For example, early recommender systems allowed users to explore a broad space of recommendations and dynamically apply different filters to narrow the space down, which gave users control over the selection of final recommendations [3]. Future work could explore how to combine computational mechanisms with user-driven filtering to enable negotiation and promote user control.

Additionally, participants noted that their needs and skills evolve over time and raised concern about outgrowing features (such as explanations) as they gained knowledge. This challenge has been echoed in prior HCI work and has been highlighted as one reason for lapsing use of informatics technologies to support self-management [17]. A flexible design approach could lend users greater autonomy to turn on or off particular features to adapt the interaction experience as their decision support needs evolve. However, while these approaches would give control directly to users it could also increase user burden and reduce their willingness to engage with technologies, as was demonstrated by research in privacy that required users to explicitly state their privacy preference [47]. Further research is needed to understand how to build flexibility into such tools in a way that does not put the burden back on the user but at the same time lends them some control to better situate a tool in the context of their lives.

5.3 Limitations

Though promising, the proposed approach has several limitations. First, with regards to the overall approach: the ability to generate highly similar meal recommendations depends on the size and diversity of the available meal catalog; if the meal catalog is limited, it could result in recommended meals having limited similarity with a reference meal. Second, this study was a preliminary feasibility investigation, with a small sample size and a simulated, lab-based setting. While the results are promising, further research is needed to validate this approach with a larger, more diverse population and in real-world environments.

6 CONCLUSION

This study demonstrates the value of similarity-based nutritional recommendations tailored to users' goals and meal plans, particularly in constrained settings. By understanding the multidimensional nature of "similarity" and opportunities to enhance user control, these findings offer important insights for designing more effective, human-centered nutrition technologies that adapt to dynamic decision-making contexts.

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